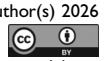
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Supportive Environmental Design for Post-stroke Rehabilitation: A Participatory Concept Mapping Study

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Abstract

Purpose: To explore home and neighborhood environment aspects that support post-stroke rehabilitation.

Background: The design of healthcare environments influences rehabilitation outcomes, however, evidence for outpatient contexts, particularly the home and neighborhood remain limited.

Method: We used descriptive statistics and cluster analysis to explore stakeholders' perspectives on environmental factors that support rehabilitation and to identify patterns in how environmental features were organized and rated.

Results: Several home and nearby environmental features were clustered around engagement in rehabilitation and participation in everyday activities. Patterns suggested that accessibility, proximity to routine services, and opportunities for social contact tended to co-occur with higher perceived support for rehabilitation.

Conclusions: The immediate environment appears to play a role in community-based recovery after stroke, with design, social context, and care processes intertwined. The findings highlight that many feasible actions such as strengthening communication, ensuring adaptive home environments, and improving neighborhood accessibility can help align housing and local planning with everyday rehabilitation after stroke. While recognizing the exploratory nature of the findings, architectural planners and healthcare practitioners could draw on these insights to inform housing planning and more inclusive environments.

Keywords

concept mapping, home rehabilitation, post-stroke recovery, home environment, neighborhood environment, participatory design

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Globally, stroke affects around 15 million people annually (Feigin et al., 2021, 2022; Thayabaranathan et al., 2022), often resulting in long-term impairments and unmet needs that limit participation in daily life (Christensen et al., 2026; Guo et al., 2021; Legg et al., 2023; Zawawi et al., 2020). Studies show that inadequate home adaptation and poor accessibility may hinder recovery and everyday participation (Elf et al., 2023; Hodson et al., 2016; Kylén et al., 2022), while supportive environments can enhance activity, engagement and rehabilitation outcomes (Anåker et al., 2017; Bernhardt et al., 2022; Killington et al., 2019; Nordin et al., 2021; Simpson et al., 2018).

In this study, we used Concept Mapping (CM) to identify and prioritize environmental features that may support home-based stroke rehabilitation, based on input from stroke survivors, family members, clinicians, and designers (Kane & Trochim, 2007; McCaffrey et al., 2019; Smith et al., 2019a, 2019b). By combining structured idea generation, sorting, and prioritization, CM offers a participatory approach for bringing together diverse perspectives and translating them into design-relevant knowledge.

The physical environment is increasingly recognized as an important determinant of health, well-being, and rehabilitation outcomes. Environmental factors such as natural light, acoustics, color schemes, and views of nature can reduce stress, improve recovery, and enhance overall quality of life in healthcare settings (Choi et al., 2012; de Lima Andrade et al., 2021; Jafarifiroozabadi et al., 2023; Li et al., 2022; Ou et al., 2024). Principles of healing and supportive environments have been increasingly integrated into hospital and institutional design (Anåker et al., 2017, 2024; Bernhardt et al., 2022; Lipson-Smith et al., 2021). However, less attention has been given to the home and neighborhood contexts in which stroke survivors continue much of their recovery and everyday life (de Vries et al., 2025; Kylén et al., 2022).

The physical environment is increasingly recognized as an important determinant of health, well-being, and rehabilitation outcomes

The design of healthcare environments is complex, involving overlapping phases of planning, design, and implementation (Elf et al., 2015). Contemporary healthcare design increasingly emphasizes evidence-informed, participatory, and user-centered approaches that engage patients, healthcare professionals, and other stakeholders (de Vries et al., 2025; Lipson-Smith et al., 2025). Evidence-based design (EBD) aims to incorporate research evidence with practical and experience-based knowledge to inform design decisions (Hamilton, 2020). However, translating diverse stakeholder perspectives into actionable design solutions remains challenging, particularly in rehabilitation settings where environmental, clinical, and everyday-life needs intersect (de Vries et al., 2025; Elf et al., 2024a; Lipson-Smith et al., 2025). Implementing supportive design in home and community care is challenging due to diverse settings, distributed responsibility, and nonstandardize routines (de Vries et al., 2025; Palinkas et al., 2025; Zhu et al., 2023).

Implementing supportive design in home and community care is challenging due to diverse settings, distributed responsibility, and nonstandardize routines

In Sweden, national stroke guidelines promote and prioritize rehabilitation in the home environment, but implementation varies across regions and municipalities (Socialstyrelsen, 2020). For environmental solutions to be feasible in practice, they need to be adapted to local contexts and developed with stakeholders who understand everyday rehabilitation work and living conditions (Harvey et al., 2023; Klaic et al., 2022; Palinkas et al., 2025). Implementation science stresses bottom-up, context-driven approaches, led by those with local knowledge (Schroeder et al., 2022).

Despite growing evidence on the role of the physical environment in stroke rehabilitation, limited knowledge exists about which home and neighborhood environmental features stakeholders perceive as important and feasible to implement. This study addresses this gap by identifying and prioritizing environmental features that may

support recovery, participation, and everyday life after stroke.

Despite growing evidence on the role of the physical environment in stroke rehabilitation, limited knowledge exists about which home and neighborhood environmental features stakeholders perceive as important and feasible to implement.

The aim of this study was to explore stakeholders' perspectives on what constitutes a supportive home and local environment for stroke rehabilitation. In addition to identifying important environmental aspects, the study also investigated how stakeholders assess the feasibility of implementing these aspects in practice.

Method

Study Design

We used CM, a participatory mixed-methods approach, to explore which environmental factors stakeholders consider most supportive for stroke rehabilitation at home (Kane & Trochim, 2007; McCaffrey et al., 2019; Smith et al., 2019a, 2019b). CM combines qualitative idea generation with quantitative analysis, enabling co-creation by stroke survivors, family members, clinicians, and designers. It captures both what people think and how they prioritize and organize ideas, making it well-suited for informing evidence-based and feasible design.

We used CM, a participatory mixed-methods approach, to explore which environmental factors stakeholders consider most supportive for stroke rehabilitation at home

The process involved four steps:

1. Idea generation—participants responded to an open-ended prompt.
2. Sorting—stakeholders grouped ideas by perceived similarity.
3. Mapping and analysis—data were analyzed using multidimensional scaling and hierarchical cluster analysis (HCA) to create a visual concept map, interpreted collaboratively.
4. Rating—each idea was rated for importance and feasibility.

Research Context and Translation

This study was conducted in Sweden with Swedish-speaking participants. All data collection, including sorting, rating, and responses, was conducted in Swedish. Contributions were translated into English by ME and MK and cross-checked using a translation tool. Where needed, translations were discussed within the research team to ensure fidelity to participants' intended meanings.

Participants and Recruitment

We invited a diverse group of stakeholders, that is, stroke survivors, family members, rehabilitation staff, healthcare managers, and architects to share insights on supportive home and neighborhood environments. Participants were recruited using convenience sampling via neurological inpatient and outpatient clinics, social media, stroke associations, and the research group's professional networks. Inclusion criteria for stroke survivors and relatives included being 18 years or older, able to give informed consent, and having direct experience with stroke. In total, 89 participants were invited; 63% were women. Inclusion criteria for rehabilitation staff and healthcare managers included having experience within stroke care, and for architects included having experience within healthcare environments.

Idea Generation

In the first CM step, participants completed the prompt "A supportive home and local environment for rehabilitation includes…" using the Sunet Survey platform (Sunet, 2024). Participants were asked to submit 2–5 responses each, resulting in 247 statements (Figure 1). The statements were initially sorted with support from IdeaDistiller, a semiautomated tool for idea

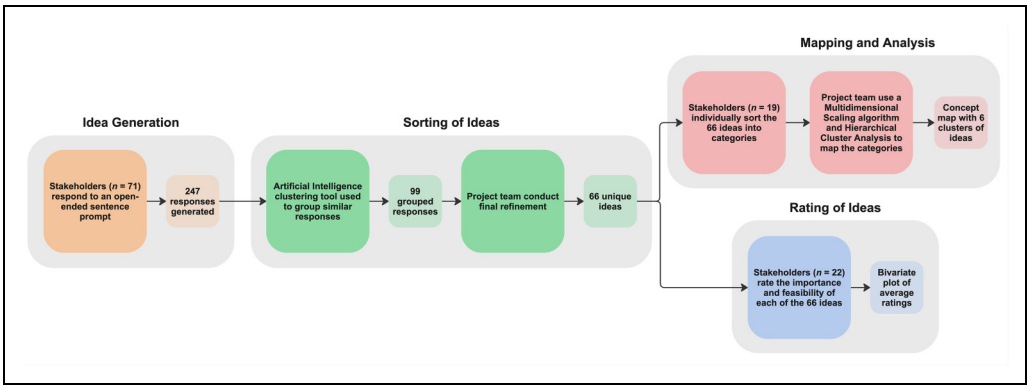


Figure 1. Concept mapping flowchart.

synthesis in concept mapping (Qwaider et al., 2026). The research team jointly reviewed and validated these groupings for thematic coherence and accuracy, and then refined the item set by removing duplicates, splitting multi-idea statements, and excluding out-of-scope content. This process resulted in 66 unique, clearly formulated ideas for the sorting phase.

Sorting and Mapping of Ideas

A subset of participants sorted 66 unique ideas using a web-based tool (Optimal Workshop, 2024), grouping them by perceived similarity and assigning labels to each category. The sorting task was completed when all ideas had been assigned to a group. Multidimensional Scaling (MDS) algorithm (Kane & Trochim, 2007; Sturrock & Rocha, 2000) was applied to visualize relationships between ideas in a two-dimensional point map. Ideas that were frequently grouped together appear closer to each other and ideas that were rarely grouped appear with more distance. A stress value of 0.217 indicated a good fit between the map and the original data (Kane & Trochim, 2007). HCA (Kane & Trochim, 2007) was used to identify conceptual groupings. The research team reviewed several candidate cluster solutions, ranging from 1 to 15 clusters, and agreed that the six-cluster solution provided the most conceptually coherent structure while avoiding both unnecessary fragmentation and oversimplification of the content.

Rating and Go-Zone Analysis

Participants then rated each idea on importance and feasibility on a 7-point Likert scale (1 = *not at all important/feasible*; 7 = *very important/feasible*). The ratings were plotted in bivariate graphs (Kane & Trochim, 2007), highlighting "go-zones" ideas rated above the mean on both dimensions.

Results

Of 89 stakeholders invited to the study, 71 (79.8%) participated in the first idea generation phase. The largest group of respondents were people with stroke ($n = 26$, 36.6%), followed by healthcare professionals ($n = 20$, 28.1%), significant others and architects ($n = 9$, 12.6% respectively), and management ($n = 7$, 9.8%; Table 1).

A total of 19 participants completed the sorting phase. The majority of these were healthcare staff ($n = 9$, 47.3%). No managers participated in this phase. Twelve stakeholders' responses were excluded due to incomplete answers. Additionally, two sorting responses were excluded: one because all ideas were placed into a single group, and another because the grouping provided no meaningful variation, making it uninformative for the analysis. For the rating phase, 22 stakeholders participated (eight people with stroke, nine healthcare staff, three architects, two significant others).

Cluster Solutions

The cluster solution was developed from the refined set of statements used in the sorting and

Table 1. Stakeholders Response Rate.

Stakeholder Group	*n* (Response Rate)
People with stroke	26 (86.7%)
Significant others	9 (75.0%)
Healthcare staff	20 (83.3%)
Management	7 (77.8%)
Architects	9 (64.3%)
Total	71 (79.8%)

rating phases. Several cluster solutions were examined, and the final solution was selected based on conceptual coherence, interpretability, and relevance to the study aim (Supplemental Material). The concept map that emerged from sorting the data revealed a rich and multidimensional view of what stakeholders consider a supportive home and neighborhood environment for stroke rehabilitation. Using MDS (Kane & Trochim, 2007), we arrived at a six-cluster solution that captured the breadth and depth of the 66 unique ideas (Figure 2). The six clusters delineate distinct yet connected dimensions of the environment: (a) A social place, (b) A place for activity and healing, (c) An enabling and adapted environment, (d) Professional competence and communication, (e) Trust and support when needed, and (f) Access to places beyond home. Together they show how place, atmosphere, adaptation, professional practice, trust, and neighborhood access jointly shape recovery at home.

Each point on the map (Figure 2) represents a single idea, and their placement reflects how often participants grouped them together, with closer points indicating stronger perceived connections.

Cluster 1—A Social Place

Cluster 1 describes the home and neighborhood as social environments that enable everyday interaction with significant others and informal contact with staff. The focus is on places and arrangements that make it easy to meet, show mutual respect, and maintain relationships, including access to nearby environments when activity at home is limited. Access to nutritious food and media technology was also mentioned as ways to sustain social connection and engagement (Figure 2).

Cluster 2—A Place for Activity and Healing

Cluster 2 describes a calm and safe home environment that supports recovery while still feeling like home. It should be fully accessible and free from excessive noise, arranged to make rest easy and everyday participation possible without becoming clinical or institutional (Figure 2). In participants' words: "A home is a home" and "A healing and harmonious environment."

Cluster 3—An Enabling and Adapted Environment

Cluster 3 focuses on the physical and technical adaptations that make rehabilitation possible in everyday life. Immediate access to appropriate home modifications, assistive technologies, and timely professional guidance was seen as essential for safe, meaningful training and for maintaining motivation to regain independence. Such adaptations help bridge the gap between clinical and home-based rehabilitation, supporting both physical and mental recovery (Figure 2).

Cluster 4—Professional Competence and Communication

Cluster 4 highlights teamwork, competence, and communication across providers and with the person with stroke, as well as coordination with authorities and employers. Participants stressed the value of staff who can support, challenge, listen, and take the person's situation into account, and the need for planned follow-ups to sustain progress.

Cluster 5—Trust and Support When Needed

Cluster 5 emphasizes receiving the right kind of support at the right time from significant others and staff, in ways that build confidence and trust. Participants highlighted the importance of clear information about available resources and barriers, knowing whom to turn to, and being able to rely on appropriate, flexible help when needed (Figure 2).

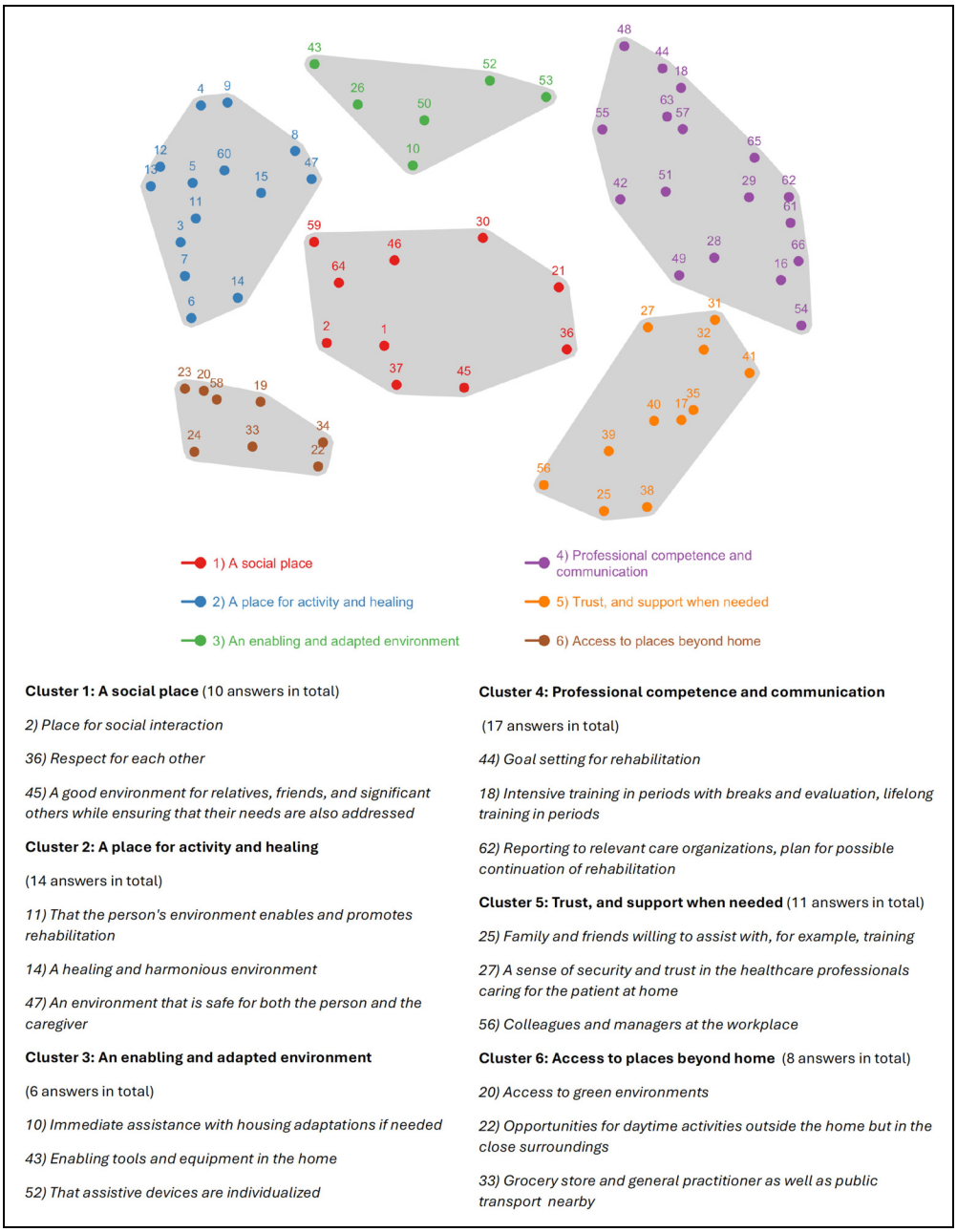


Figure 2. Final concept map showing the prompt used and each of the 66 ideas as numbered dots in the map, positioned through multidimensional scaling of the sorting data. The six shaded areas show the identified concepts achieved by hierarchical cluster analysis. The six clusters are represented with the number of answers and examples of ideas.

Cluster 6—Access to Places Beyond Home

Cluster 6 recognizes the importance of easy access to nature and everyday meaningful destinations. Safe and accessible surroundings that enable outdoor rehabilitation, together with nearby shops, healthcare, and public transport,

were considered necessary to maintain activity outside the home (Figure 2).

Go-Zone Analysis

Figure 3 presents a bivariate plot where each of the 66 unique stakeholder ideas is positioned according to the average rating for importance (*x*-axis) and feasibility (*y*-axis). The upper right-hand corner of the plot, known as the "go-zone," contains 20% of the ideas that got the highest ratings on both dimensions. In this study, the go-zone threshold was set at >6.55 for importance and >5.36 for feasibility. A total of nine ideas fell within the go-zone (Table 2). These included ideas from four of the six clusters, with the majority belonging to Cluster 4—Professional Competence and Communication and Cluster 3—An Enabling and Adapted Environment. The highest-rated item in importance (6.91) and high feasibility (5.73) was "to listen to and take the person into account." Other go-zone items included "goal setting for rehabilitation," "that assistive devices are individualized," and "an environment that is safe for both the person and the caregiver." Overall, ideas from Cluster 4—Professional Competence and Communication received the highest average scores across dimensions (importance and feasibility), as shown in the ladder graph (Figure 4). This cluster reflects stakeholder emphasis on communication, stroke-specific competence, and coordination between care providers.

Discussion

This study identified six themes describing supportive home and neighborhood environments for stroke rehabilitation. Together, these themes reveal that effective home-based rehabilitation relies on the interplay of spatial, relational, and organizational factors, rather than on physical design alone. The environment must be understood, designed, and used as an active component of the rehabilitation process. This requires moving beyond the notion of the environment as neutral and toward an understanding of the environment as a dynamic factor that can support or hinder recovery (Bollinger et al., 2024; Chen et al., 2024; Elf et al., 2024b; OECD, 2023; Qin et al., 2022).

This study identified six themes describing supportive home and neighborhood environments for stroke rehabilitation. Together, these themes reveal that effective home-based rehabilitation relies on the interplay of spatial, relational, and organizational factors, rather than on physical design alone.

Designing for home-based rehabilitation should therefore integrate these dimensions rather than treat them separately. Using CM made it possible to integrate diverse perspectives from stroke survivors, relatives, and professionals, highlighting how supportive environments are understood in practice. As the method captures perceived relationships rather than causal ones, the resulting map reflects shared meanings and priorities rather than measured effects.

The go-zone analysis reinforced this picture by highlighting stakeholder priorities rated both important and feasible. Most of the highly rated ideas concerned relational and adaptive aspects such as professional competence, respectful communication, individualized support, and planned follow-up, rather than fixed environmental features; these findings align with core elements of relational and person-centered care (Martín-Sanz et al., 2022; McCance & McCormack, 2017; McCormack et al., 2021; Santana et al., 2018; WHO, 2013). Accordingly, design processes should translate these factors into concrete requirements, for example, spaces that enable private yet collaborative encounters, technologies that support follow-up and feedback, and care routines that ensure continuity and responsiveness at home.

In practice, participation, continuity, and clear communication should be the default. Systems and physical settings need to be organized to sustain interprofessional competence and timely responses, for example through accessible information flows, coordinated home visits, and environments that facilitate shared decision-making between staff, stroke survivors, and family

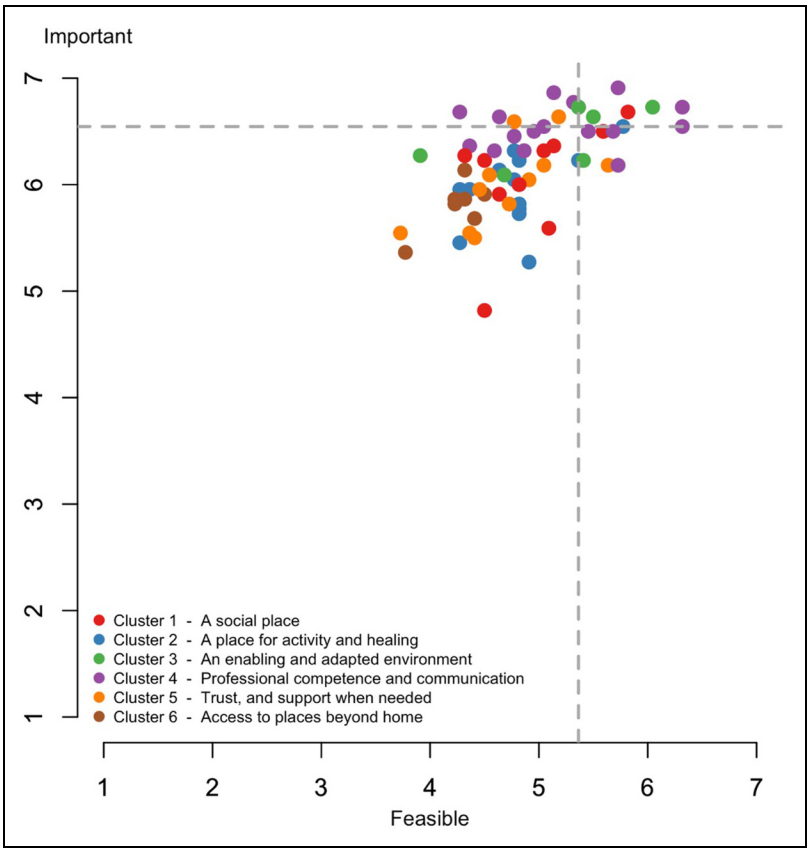


Figure 3. The go-zone map, a bivariate plot, shows the stakeholders ratings for each of the 66 ideas where the *x*-axis represents importance and *y*-axis represents feasibility. The go-zone is shown in the upper right-hand quadrant.

members (Martín-Sanz et al., 2022; McCance & McCormack, 2017; McCormack et al., 2021; Santana et al., 2018; WHO, 2013).

Participants emphasized the need for adaptive environments, individualized aids, and safety supported by planned follow-up. Prior work also shows that people rehabilitating at home seek adaptations and safety in everyday contexts (de Vries et al., 2025; Elf et al., 2023; Kylén et al., 2022). Together, these findings suggest that "supportive environments" are defined less by discrete physical elements and more by what those elements enable: motivation, autonomy, participation, relationships, and continuity of care over time. Building on these insights, future work should examine how such priorities can be realized in practice through participatory design and implementation studies. Testing concrete approaches, for example, home modification protocols, digital coordination tools, personalized assistive devices, or neighborhood-based rehabilitation hubs could clarify how feasibility and impact align in real-world contexts.

Interprofessional communication, colocation, and coordinated teamwork lead to better outcomes in hospital-based stroke care (Langhorne et al., 2020), and ward layout can enable or hinder collaboration (Anåker et al., 2017; Paxino et al., 2022). The challenge for home-based rehabilitation is to achieve equivalent coordination without physical colocation. Our findings point to technology-enabled infrastructure and simple, reliable procedures as practical enablers when proximity is not possible. Centralized clinical

Table 2. Ideas in the Go-Zone.

Cluster	No.	Item	Importance	Feasible
Cluster 4 Professional competence and communication	55	To listen to and take the person into account	6.91	5.73
Cluster 4 Professional competence and communication	51	Information about assistive devices	6.73	6.32
Cluster 3 An enabling and adapted environment	52	That assistive devices are individualized	6.73	6.05
Cluster 4 Professional competence and communication	48	Understanding that it takes a long time, possibly life-long consequences	6.73	5.36
Cluster 3 An enabling and adapted environment	50	That there is access to the necessary aids	6.73	5.36
Cluster 1 A social place	36	Respect for each other	6.68	5.82
Cluster 3 An enabling and adapted environment	53	The motivation and challenge of improving both physical and mental well-being	6.64	5.50
Cluster 4 Professional competence and communication	44	Goal setting for rehabilitation	6.55	6.32
Cluster 2 A place for activity and healing	47	An environment that is safe for both person and the caregiver	6.55	5.77

According to the average rating on importance and feasibility on a 7-point Likert scale (1 = *not at all important/feasible*; 7 = *very important/feasible*).

records, videoconferencing, and other tools may support professional collaboration, while easy, consistent two-way communication with people with stroke is essential for trust, shared decision-making, and a continuous experience of follow-up.

Accessible homes and public spaces that support social connection are necessary, but stakeholders also emphasized cultural inclusion, such as awareness, respect, and listening, highlighting the broader social context of care. The ideas resonate with work on dementia-friendly communities, where community education complements inclusive design and planning (de Vries et al., 2025; Shannon et al., 2019). For local communities to be truly supportive, opportunities to inform and educate the wider public about stroke and rehabilitation are needed. "Access to places beyond home" was rated least feasible, likely reflecting perceived geographic and planning constraints. Rather than treating this as a fixed limitation, we recommend designing, prototyping, and evaluating hybrid pathways to determine which needs can be met virtually (telehealth, online community resources) and which require physical proximity via local hubs, outreach, or transport. Clarifying what can be done virtually versus in person can help target investments across diverse communities and locations. These findings also point to the importance of cultural shift and community-level design and policy. Accessibility, public transport, and social inclusion are not individual challenges but collective ones, requiring coordination between health services, housing, and urban planning.

Overall, EBD for home-based stroke rehabilitation should integrate relational and adaptive considerations with spatial ones. The home is not merely a functional container; it is tied to identity, ownership, and autonomy (Meijering et al., 2016). Designing for impact therefore means building relational capacity and simple systems so that home and neighborhood enable participation, trust, and control and provide place-based supports when proximity is required.

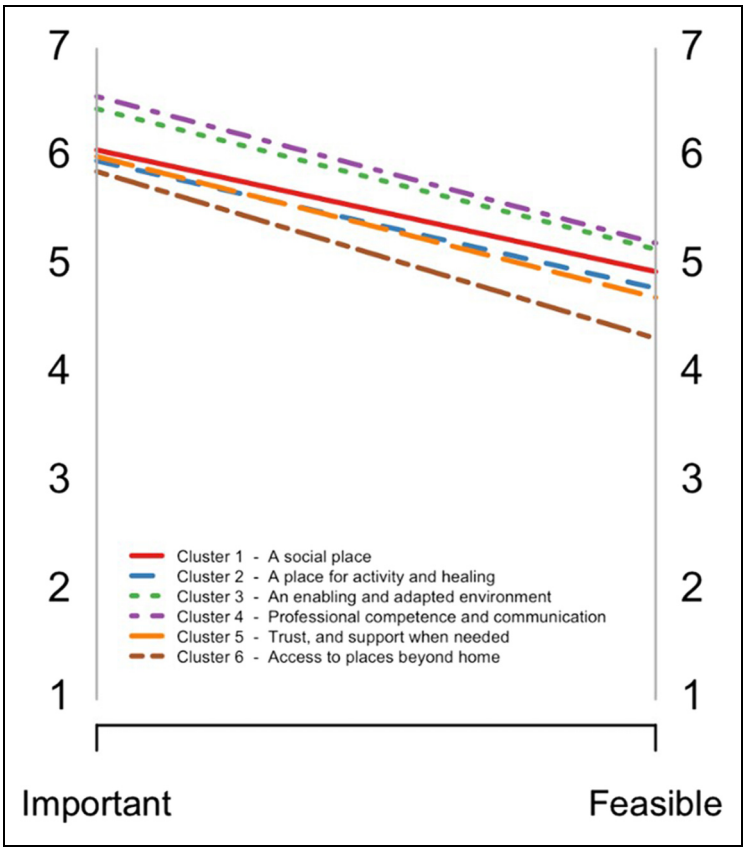


Figure 4. Ladder graph showing overall importance and feasibility per cluster.

Strengths and Limitations

While this study was conducted within the Swedish healthcare and housing context, the themes identified, especially the interplay of communication, adaptation, and accessibility, are likely transferable to other countries and contexts where home-based rehabilitation is becoming increasingly common. The participatory approach, which brought together stroke survivors, relatives, and professionals, is a particular strength, grounding the findings in diverse real-world perspectives. The specific pathways to achieve these priorities, however, will depend on local infrastructure and policy frameworks. Stakeholder input was elicited through a single open-ended prompt, enabling a bottom-up process grounded in lived experience and practice. This inclusiveness means the results mirror participant priorities rather than predefined categories. The go-zone output provides an actionable starting point for aligning design features with real-world needs; future work should unpack feasibility (e.g., cost, time, complexity) for specific ideas to understand how their implementation could be supported. We observed that items rated as more important also tended to be rated as more feasible; this may reflect desirability bias in individual judgments, although the independent rating used in this study (without group discussion) could have tempered such effects. Participation was broad, with strong representation from people with stroke, but healthcare managers did not take part in sorting or feasibility rating. Future CM studies should include implementation-critical stakeholders (e.g., managers, policy makers, planners). Using an AI tool in sorting the generated unique ideas was helpful as it assisted the research

team in grouping the data; manually checking the groupings generated by the AI was more time-effective than manually categorizing the unsorted data (Singh et al., 2024).

Conclusion

A whole-system approach is needed to support stroke rehabilitation at home and community care across disciplines, an informed wider society and inclusive environments. Many identified actions were considered feasible to implement, especially those relating to professional competence and communication, and enabling environments. These findings point to near-term opportunities to better align housing, neighborhood design, and service provision with rehabilitation and everyday life after stroke.

Implications for Practice

- Home rehabilitation is related to care processes and the home's design, and these aspects cannot be treated as separate considerations.
- Spatial, social, and organizational conditions interact to shape rehabilitation outcomes.
- As home-based care expands, designers and healthcare professionals should work jointly using participatory, iterative methods to carry relevant ideas through to implementation.
- CM can act as a shared tool to translate lived experience into design briefs and service improvements, supporting collaboration from planning through evaluation.

Acknowledgments

We sincerely thank all participants who generously shared their experiences and perspectives, contributing to a deeper understanding of what makes home and neighborhood environments supportive for stroke rehabilitation. We would also like to acknowledge statistician Katrin Gunnarsdottir from Regional Cancer Center West, Gothenburg, Sweden, for supporting us with the analytical stage of the CM data.

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Ethical Considerations

This study received approval from the Swedish Ethical Review Authority (Dnr 2022-04231-01). Informed consent was acquired from participants prior to the study. Written information on the study purpose and what would be involved in participating was given before informed consent.

Authors' Contributions

Study design: MK, ME, FS, and RLS; acquisition of data: LdV; analysis and interpretation of data: MK, ME, LdV, and FS; drafting of the manuscript: ME, MK, and RLS; critical revision of the manuscript for important intellectual content: ME, MK, RLS, FS, HPR, and LdV. All authors read and approved the final manuscript.

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Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request and anonymized before dissemination.

Supplemental Material

Supplemental material for this article is available online.

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